***Brock Biology of Microorganisms, 15e* (Madigan et al.)**
Chapter 29   Epidemiology

29.1   Multiple Choice Questions

1) A disease that is present in unusually high numbers throughout the world is called a(n)
A) endemic.
B) epidemic.
C) sporadic.
D) pandemic.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

2) A **________** is a disease that primarily infects animals but can be transmitted to humans.
A) nosocomial infection
B) zoonosis
C) vector infection
D) mudurane
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

3) An inanimate object that transmits infectious agents between hosts is most appropriately called a
A) fomite.
B) carrier.
C) vector.
D) reservoir.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

4) Which of the following is NOT a public health measure used to control the transmission of disease?
A) sanitary water and waste disposal methods
B) immunization
C) forced quarantine
D) genetic engineering
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.5

5) Which of the following is/are considered (a) direct contact infection(s)?
A) syphilis only
B) gonorrhea only
C) skin infections only
D) syphilis, gonorrhea, and skin infections
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

6) The onset of a given epidemic is indicated by a sharp rise in the number of cases reported daily over a brief interval. This indicates that the mode of transmission is
A) host to host.
B) a common source.
C) insect vector.
D) mechanical vector.
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.4

7) Which of the following diseases is NOT normally spread by a common source?
A) measles
B) foodborne diseases
C) waterborne diseases
D) cholera
Answer: A
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.4

8) Which of the following is an example of herd immunity?
A) Brucellosis is no longer found in farm animals in the United States.
B) If 70% of the population is immunized against polio, the disease will be essentially absent from the population.
C) Federal law requires that all cattle not immune to anthrax be destroyed.
D) All farm animals used for food must be immunized against all the common agents of disease that infect humans.
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.2

9) Potential candidates for biological warfare
A) are generally gram-negative rather than gram-positive.
B) can be virtually any pathogenic bacterium or virus.
C) are eukaryotic rather than prokaryotic.
D) must be genetically engineered to be effective.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.9

10) Based on knowing the history of the HIV/AIDS pandemic, and of efforts to track its spread, you can conclude that it is likely that the incidence of HIV/AIDS among transfusion recipients has
A) increased since the discovery of HIV.
B) decreased since the discovery of HIV.
C) remained the same since the discovery of HIV.
D) never been more than a few individuals per year.
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.8

11) Which of the following shows the correct relationship among the epidemiology terms listed?
A) prevalence > incidence > mortality
B) incidence > prevalence > mortality
C) mortality > morbidity > prevalence
D) mortality > incidence > prevalence
Answer: A
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.1

12) Which of the following emergence factors has contributed to the HIV/AIDS epidemic?
A) climate change
B) rapid pathogen adaptation and change
C) increasing travel to endemic areas
D) exotic pet and meat trade
Answer: B
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.7

13) *Treponema pallidum* is extremely sensitive to temperature changes and low moisture, thus it is transmitted
A) by intimate person-to-person contact.
B) by fomites.
C) by vectors.
D) through common sources such as food and water.
Answer: A
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.3

14) The most common vector-borne disease in the United States is
A) influenza A.
B) Lyme disease.
C) malaria.
D) pneumonia.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.7

15) A large number of cases of a particular disease observed in a relatively short period of time in an area that previously experienced only sporadic cases of the disease is known as a(n)
A) pandemic.
B) outbreak.
C) endemic.
D) zoonosis.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

16) Which stage of an acute infectious disease occurs between the time the organism begins to grow in the host and the appearance of disease symptoms?
A) acute period
B) decline period
C) infection
D) incubation period
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

17) A marked seasonality to a disease is often indicative of
A) certain modes of transmission.
B) the presence of carriers.
C) a zoonotic infection.
D) a bacterial disease.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.2

18) Which of the following are NOT vectors important in disease transmission?
A) fomites
B) insects
C) ticks
D) rodents
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

19) An example of a cyclical disease is
A) diphtheria.
B) smallpox.
C) California encephalitis.
D) anthrax.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

20) Which body site is preferentially infected by foodborne pathogens?
A) gastrointestinal tract
B) respiratory tract
C) cerebrospinal fluid
D) liver
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.4

21) Influenza pandemics occur cyclically because
A) less than 100% of the population is immunized.
B) new strains emerge due to reassortment between bird, swine, and human variants.
C) the vector is seasonal.
D) there are environmental reservoirs that release the virus during particular seasons.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.8

22) Cholera is an example of a pandemic disease that
A) has multiple wild animal reservoirs and is thus difficult to eradicate.
B) spreads from a common source and could be controlled with adequate clean water and waste sanitation measures.
C) is spread through direct contact and has only a human reservoir.
D) has recently emerged due to overcrowding in urban centers.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.8

23) Ebola is caused by
A) *Streptococcus pneumoniae*.
B) the Zika virus.
C) a filovirus.
D) a rhinovirus.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.7

24) What is a/are potential reservoir(s) for the rabies virus?
A) water
B) insects
C) bats
D) fomites
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

25) Which disease listed does NOT require quarantine?
A) smallpox
B) HIV/AIDS
C) plague
D) cholera
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.5

26) Filoviruses (such as Ebola virus) that cause severe hemorrhagic fevers generally have high ________ but low ________.
A) incidence / prevalence
B) prevalence / mortality
C) mortality / morbidity
D) morbidity / mortality
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.1

27) Which group of organisms is difficult to control through immunization because of their rapid and unpredictable genetic mutations?
A) vector-borne organisms
B) RNA viruses
C) *Archaea*
D) DNA viruses
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.7

28) Which public health measure(s) is/are most effective against pathogens transmitted through common vehicles?
A) water purification
B) mosquito control
C) food safety regulations
D) water purification and food safety regulations
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.5

29) Which of the following diseases would be the easiest to control in a human population?
A) an infectious disease with wild animals as a reservoir
B) an infectious disease with humans as the only reservoir
C) an infectious disease with domestic cows as the only reservoir
D) an infectious disease with several possible reservoirs
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.5

30) The number of ill individuals within a population is referred to as
A) morbidity.
B) mortality.
C) residency.
D) prevalence.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

31) Disease ________ is measured by the total number of new reported disease cases within a population over a period of time.
A) incidence
B) frequency
C) morbidity
D) prevalence
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

32) Influenza pandemics occur periodically due to characteristics of the influenza virus. How did these characteristics, in this case of the H1N1 strain, contribute to the spread of the swine flu pandemic in 2009?
A) The H1N1 virus had a mutation that increased the mortality associated with infection.
B) The H1N1 virus had a mutation that increased the basic reproduction number of the virus.
C) The H1N1 virus underwent a significant antigenic shift compared to other circulating strains of influenza.
D) The H1N1 virus acquired a new gene from pigs that resulted in increased virulence.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.8

33) A nonliving source of an infectious agent that infects a large number of people is called a
A) fomite.
B) reservoir.
C) vector.
D) vehicle.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

34) The field of study that specifically examines the occurrence, distribution, and determinants of health and disease in a population is
A) microbiology.
B) immunology.
C) epidemiology.
D) virology.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

35) People who have a subclinical case of a disease are frequently ________ of a particular disease.
A) fomites
B) carriers
C) vectors
D) vehicles
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.1

36) With regards to mode of disease transmission, respiratory pathogens are generally ________, and intestinal pathogens are generally spread by contaminated ________.
A) spread by direct contact / vectors
B) spread by indirect contact / carriers
C) more transmissible / needles
D) airborne / food or water
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.3

37) A disease showing a relatively slow, progressive rise followed by a gradual decline in incidence is indicative of a(n)
A) host-to-host epidemic.
B) common source epidemic.
C) biological weapon.
D) endemic disease.
Answer: A
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.4

38) Diseases that are good candidates for eradication have
A) only a human reservoir.
B) no asymptomatic phase.
C) an environmental reservoir.
D) only a human reservoir and no asymptomatic phase.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.5

39) A disease present at a constant low level in a population is described as
A) zoonotic.
B) common source.
C) endemic.
D) epidemic.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

40) Diseases that suddenly become more prevalent are referred to as ________ diseases.
A) indirect
B) emerging
C) vector
D) common source
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.7

41) Pasteurization of milk is a process that reduces bacterial levels and is an example of a disease control measure aimed at
A) preventing host-to-host transmission.
B) controlling the disease vector.
C) preventing common source diseases.
D) eliminating the disease reservoir.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.4

42) ″Typhoid Mary″ is an example of a ________, because she was infected by the causative agent for typhoid fever yet was asymptomatic.
A) fomite
B) carrier
C) vector
D) vehicle
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.3

43) Most cases of mosquito-transmitted diseases occur in
A) the summer and fall seasons.
B) tropical and sub-tropical regions.
C) rural areas.
D) tropical and sub-tropical regions or during the summer and fall seasons.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

44) Over time, the relationship between a pathogen and a naïve susceptible population tends towards
A) the extinction of the host.
B) the extinction of the pathogen.
C) a balance between host and pathogen such that both are maintained.
D) the extinction of either the host or the pathogen.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.2

45) The most recent influenza pandemic was in
A) 1918.
B) 1961.
C) 1979.
D) 2009.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.8

46) Epidemiological studies of AIDS suggest that
A) socioeconomic factors may affect infection risk.
B) the incidence is increasing in recent years.
C) heterosexual women are at lower risk of infection than other women.
D) the virus is no longer changing and evolving.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 28.9

47) The basic reproduction number of a pathogen
A) can vary during an epidemic based on infection control measures.
B) can be determined experimentally in the lab.
C) is usually high for pathogens that are transmitted through direct contact.
D) can be determined experimentally and is usually high for pathogens that are transmitted through direct contact.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.4

48) In just over a decade of efforts, the World Health Organization (WHO) eradicated smallpox by using worldwide
A) education campaigns.
B) destruction of infected domestic animals.
C) education campaigns and destruction of infected domestic animals.
D) vaccination programs.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.9

49) Diseases can be controlled through immunization even if the percentage of the population that is immunized is less than 100% because
A) vectors can be controlled through other means.
B) most diseases lack reservoirs.
C) of herd immunity.
D) vehicles can be sterilized.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.2

50) The term ________ is applied to strains and preparations of *Bacillus anthracis* that exhibit properties that enhance dissemination and use as biological weapons.
A) weaponized
B) virulized
C) category A
D) infectious
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.9

29.2 True/False Questions

1) A disease that is constantly present in low numbers is called an acute disease.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

2) Since 1994, HIV is no longer more prevalent in certain groups of people.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.8

3) Many pathogenic organisms require living hosts as reservoirs to survive.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

4) Upper respiratory infectious agents are commonly transmitted from person to person.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

5) Morbidity statistics more precisely define the health of a population than mortality statistics.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

6) In the absence of susceptible hosts, *Clostridium tetani* would still survive in nature.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.2

7) In the United States and other developed countries, deaths due to infectious diseases are decreasing.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

8) An endemic disease is constantly present, usually at low incidence, in a population.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

9) Reservoirs of infectious disease agents may be either animate or inanimate.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

10) When the life cycle of a disease agent is dependent on a single host species, the pathogen can be eradicated.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.5

11) Food and water are considered disease vehicles.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

12) Failure to reach equilibrium with a disease agent could result in extinction for a host species.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.2

13) Many parasites, such as those that cause malaria, use antigenic variation to decrease virulence within a specific host.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.2

14) Immunization against endemic diseases is NOT necessary when traveling outside of one′s home country or region.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.6

15) Emerging infectious disease will likely affect only developing countries in the near future.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.7

16) Changes in food processing and distribution can increase the incidence of new and emerging diseases.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.7

17) To control a disease in a population, 100% immunization is necessary.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.3

18) The lower the basic reproduction number of a pathogen, the higher the percentage of immune individuals necessary to provide herd immunity.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.4

19) A disease that is transferred by direct intimate contact with a mortality rate over 90% would be an effective biological weapon.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.9

20) A disease transferred by indirect contact with a high basic reproduction number and mortality would be an effective biological weapon.
Answer: TRUE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 29.9

29.3 Essay Questions

1) Give an example of an acute and a chronic carrier of disease. What disease characteristics allow for chronic carriers to play significant roles in the spread of the disease? How are diseases with chronic carriers controlled?
Answer: Acute disease carriers contain only the infectious agent for an ephemeral time period, such as with a cold or flu virus, whereas chronic individuals carry the infectious agent long-term, such as with HIV/AIDS or tuberculosis. The disease characteristic that allows for chronic carriers to be important in spreading a disease is a very long incubation period in which the carrier is infected but has relatively few symptoms. The longer the incubation period, the more that could be infected by the chronic carrier. One way to control the spread of a disease with a long incubation time is to screen the population (or any at risk population) for the disease even if they lack symptoms. HIV testing during routine exams and the tuberculin skin test are examples.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.1

2) Why are diseases such as rabies that have a wild animal reservoir so difficult to eradicate while other diseases such as brucellosis that reside in domestic animals have been virtually eradicated.
Answer: The eradication of a disease requires that the pathogen be no longer transmitted. Because transmission of a disease such as rabies comes from wild animals that serve as reservoirs, all reservoirs would need to be removed for there to be no more transmission of the infectious agent. Immunizing or destroying all wild animals is not practical, so rabies remains very difficult to eradicate. Brucellosis and many other diseases are more easily eradicated because the reservoir is more easily controlled when, for example, they occur primarily in domesticated animals that can be tested, immunized, and or destroyed if infected.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 29.5

3) What measures would be taken if a case of smallpox were recognized in a world traveler after the individual had debarked from Heathrow airport in England following several days of sightseeing in London?
Answer: The World Health Organization (WHO) would first quarantine the traveler with the smallpox disease to avoid any further spread of the infectious agent. Next, all individuals known to have contact with the infected individual would be quarantined, observed, and immunized to ensure they would not become ill. The WHO could also vaccinate people who might not have had direct contact with the infected person but were present at the outbreak site. Very few people are immunized against smallpox in the current population, thus almost the entire world population is susceptible. The WHO would likely take a very aggressive and proactive approach to ensure that smallpox would not spread, given its high mortality rate and the extremely large susceptible population.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.6

4) You are planning a whitewater rafting vacation in Kenya. Based on what you know about microbiology, what precautions should you take to return in the best of health?
Answer: Caution should be taken to (1) not transfer any infectious agent to Kenya and (2) avoid becoming infected by a pathogen while in Kenya. A review on immunization status and medical records through consultation with an appropriate clinician should help to determine if one is ill and provide any vaccinations that are necessary. For example, an individual might be up-to-date on all necessary vaccinations for travel in the United States, but Kenya could have other infectious agents that Americans are not normally immunized against, such as typhoid and yellow fever. The CDC Information for Travelers also provides information regarding health concerns for international travel.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 29.6

5) How can epidemiologist differentiate between common-source and host-to-host epidemics based only on the incidence of disease?
Answer: Common-source epidemics result from a single contaminated source, which can be transmitted in any possible manner. Host-to-host epidemics result from person-to-person contact. Common-source epidemics usually involve a large population of individuals suddenly becoming ill, and host-to-host epidemics generally have a more gradual increase and decrease in the number of diseased individuals.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 29.3

6) Explain the concept of coevolution in terms of host-pathogen interactions.
Answer: The host and pathogen both evolve and affect each other. As the pathogen evolves virulence factors, a host evolves defenses against them. This coevolution drives the establishment of a balanced equilibrium, which favors the survival of both the host and the pathogen. If this balance is not obtained, the host could be killed due to the infectious agent′s pathogenicity, which would also in turn result in death of the pathogen as well.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 29.2

7) List and discuss five factors responsible for the emergence of new pathogens or the reemergence of existing pathogens.
Answer: Any of the following seven are possible answers: breakdown of public health standards due to war, migration, or natural disasters; catastrophic events that upset the usual host-pathogen balance; changes in land use; changes in human demographics and behavior; increased international travel and commerce; antimicrobial resistance, and climate change affecting vector ranges.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.7

8) Rodents have been associated with numerous severe epidemics in the past, prompting microbiologist Hans Zinsser in the 1930s to write a book, *Rats, Lice, and History*, in which he documented the important role of rodents in epidemics. Discuss why rodent control is still desirable, even in an age when it is perceived that the ″magic bullets″ of chemotherapy and immunization to control infectious disease are readily available.
Answer: Rodents, among other animals, can serve as reservoirs and vectors of infectious diseases. Due to rodents′ high reproductive rate, they can quickly propagate infectious diseases and in little time engender disease epidemics. Even if a disease is easily curable with a particular drug, it is still desirable to consider upstream efforts to eliminate the reservoir of an infectious disease, thus reducing human exposure to the disease agent. Currently emerging diseases such as hanta virus have a rodent reservoir and both plague and typhus can still cause significant numbers of cases worldwide.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 29.6

9) How might climate change impact the incidence of a disease? Use a specific example to illustrate your answer.
Answer: Climate change could increase temperatures in moderate climates, allowing vectors or other animals to expand their range. If climate change results in the movement of animals into areas where people who have not been around those animals before, zoonotic and/or vector-borne diseases could increase. For instance, warmer temperatures and increased rainfall in Florida might allow certain tropical mosquitos to survive and reproduce, thus spreading tropical vector-borne diseases such as malaria, dengue fever, or yellow fever. A sudden increase in the number of mosquitoes, which serve as excellent vectors to transmit infectious agents, can cause a major increase in the incidents of diseases such as dengue fever, malaria, Rift Valley fever, West Nile encephalitis, yellow fever, etc. Movement of wild mammals into closer contact with humans would be another example of how climate change could impact the incidence of a disease.
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 29.7

10) What are the five stages of a typical infectious disease? During which stage is the host likely to infect others?
Answer: Infection occurs when the infectious agent invades and propagates in the host. The incubation period occurs just after infection and includes the time when the pathogen has yet to cause disease as well as cause the first (weak) symptoms. The acute period occurs when the disease symptoms are at its peak, and once the symptoms begin to subside it is considered the decline period. Convalescence is the last period of disease progression where the individual regains strength and returns to normal health. The incubation period is the stage in which the host is most likely to infect others, although hosts can often still infect others during the acute period as well. Convalescence is the stage in which many hosts will no longer be infectious.
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 29.1

11) Why are public health officials concerned about smallpox being used as a biological weapon? Give several reasons in your answer.
Answer: The causative agent for smallpox, *Variola major*, kills approximately 30% of those individuals who are infected with the virus. Along with a high mortality rate, the virus is rapidly disseminated by aerosol spray. The virus can also spread within a population through person-to-person contact. Because over 90% of people worldwide are susceptible to smallpox, the concern for the virus being used as a biological weapon is great.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 29.8